

Question	Answer	Marks
10(a)	• <u>redshift</u> is the increase in observed wavelength / decrease in observed frequency (caused by Doppler effect) • radiation from distant galaxies is observed to be redshifted • redshift provides evidence that galaxies are moving apart • <u>galaxies</u> moving apart means Universe must be expanding Any three points, 1 mark each	B3
10(b)(i)	$F = L / 4\pi d^2$	C1
	$d = \sqrt{(1.90 \times 10^{36} / [4\pi \times 8.42 \times 10^{-16}])}$ $= 1.34 \times 10^{25} \text{ m}$	A1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $(726 - 658) / 658 = v / (3.00 \times 10^8)$	C1
	$v = 3.1 \times 10^7 \text{ m s}^{-1}$	A1
10(c)(i)	line with positive gradient passing through the origin	B1
	straight line with positive gradient	B1
10(c)(ii)	Hubble constant	B1